

Data Engineering and Machine Learning Pipeline

Project Report

ENGE707 Data Engineering and AI

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1. Problem Definition and Dataset Selection

1.1 Problem Statement

There are countless factors in people's daily lives which can influence their health, health decisions such as sleep, physical activity, diet, alcohol intake and other lifestyle habits. Being able to understand how these factors could influence serious health conditions could help people make better lifestyle choices in regards to their health. Heart disease is the biggest global health problem, causing 19.8 million deaths in 2022 alone. Being able to identify lifestyle factors that are associated with heart disease could help people make more cautious health choices, before heart disease develops.

Currently, identification of heart disease requires a major cost as in New Zealand, it approximately cost 538.8 million NZD (Telfar Barnard, 2025). Unaffordability is one of the main causes as well as geographically, and the ability to receive frequent testing for the general population. Even though the primary affected group is the general population, the health care facilities and personnel are also affected. This is because if the government would like to encourage more people to get regular health checks, this would overwhelm the facility thus causing less possibility for individuals that actually need testing. Figuring out the lifestyle that tends to give a higher chance of heart disease by the use of a model would increase its availability to determine whether further actions are needed.

The problem investigated in this project is whether the information of patients such as vitals and lifestyle choices could be used to predict whether or not they have heart disease. Instead of one characteristic, this project will investigate how multiple patient characteristics relate to heart disease and determine which of these variables are useful for accurately predicting heart disease.

The analytical goal of this project is to develop a machine learning model that can predict whether a person belongs to the heart disease category using relevant patient information from data analysis. This would benefit the stakeholders of our project, these being the **general population, healthcare workers, and researchers.**

1.2 Dataset Selection

The dataset we selected for this project is the “*281K Patient Dataset for Multi-Disease Prediction*” sourced from Kaggle on 18 August 2026 and is released under the CCO Public Domain licence.

<https://www.kaggle.com/datasets/danishjmeo/281k-patient-dataset-for-multi-disease-prediction?resource=download>

This dataset contains 280,985 patient records and 39 columns, which include clinically recorded vitals, lifestyle choices, demographic information, and documented diseases. This makes the dataset suitable for our problem statement as it contains all these measurements which can be analysed to form a possible prediction of potential heart disease.

This integrated dataset is a combination of 3 Kaggle datasets that are available publicly. The 3 datasets cover diabetes, heart disease, and hypertension. This source credits the 3 datasets mentioned

(<https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset>,

[kaggle.com/datasets/oktayrdeki/heart-disease](https://www.kaggle.com/datasets/oktayrdeki/heart-disease),

[kaggle.com/datasets/ankushpanday1/hypertension-risk-prediction-dataset](https://www.kaggle.com/datasets/ankushpanday1/hypertension-risk-prediction-dataset))

2. Data Acquisition, Inspection, and Documentation

2.1 Data Acquisition and Dataset Structure

The dataset was loaded with the Pandas library in python, and we saved a copy of the dataset before any cleansing or transforming was done. During the initial inspection, python returned **280985 records** and **39 columns**. Python also showed that the data had 3 data types: 19 float columns, 1 int column, and 19 string columns.

Each of the columns describe a different information on each patient, however the main ones can be categorised into demographic, lifestyle, clinical measurements, and disease related information.

Variable Type	Features and Meanings
Demographic	• age, Patient age in years • age_level, Age level is the age group • age_normalized, Normalized age (0,1) • gender, Gender • education_level, Education level • employment_status, Employment status
Lifestyle	• smoking, Whether the instance smokes or not • sleep_hours, Average daily sleep in hours • physical_activity, Physical activity level • stress_level, Stress level • sugar_consumption, Daily sugar intake level • alcohol_intake, Alcohol consumption in numeric score • salt_intake, Dietary salt intake in numeric score
Clinical	• bmi, BMI (kg/m2) • bmi_level, BMI category (WHO standard) • hbA1c_level, Long-term glucose indication in percentage • glucose, Blood glucose level

	• cholesterol, Total cholesterol level • triglycerides, Triglyceride level • blood_pressure, Blood pressure (mmHg) • crp_level, CRP level • homocysteine_level, Homocysteine level • systolic_bp, Systolic BP • diastolic_bp, Diastolic BP • heart_rate, Resting heart rate (bpm) • hdl, HDL cholesterol level • ldl, LDL cholesterol level
Disease Indicators	• diabetes, Whether the instance has diabetes or not • hypertension, Hypertension indicator • heart_disease, Heart disease indicator, how common heart disease is for a patient with a similar record. • family_history, Family history of chronic disease • low_hdl_cholesterol, Low HDL indicator • high_ldl_cholesterol, High LDL indicator • high_blood_pressure, High BP indicator
Others	• composite_key, Identifier that represents a demographic profile across (age group, gender, BMI category, smoking status, diabetes status combination) • source_dataset, Original source dataset • disease_flags, Disease combination code • sublabel, Classifies the exact kind of disease a person has • label, Binary target, classifies if a person has a disease or not.

2.2 Initial Data Quality Inspection

The raw dataset was inspected for missing values, duplicate records, unique values, categorical inconsistencies, numerical ranges, and unusual values before any cleaning was performed.

2.2.1 Missing Values

```
# Missing values
missing_values = df_original.isnull().sum()
missing_values[missing_values > 0]
```

✓ 0.4s

Series([], dtype: int64)

This screenshot shows that 0 values are missing in the 280985 instances in the dataset. It is highly unusual for a dataset of this size to be completely filled with values, and from the dataset, the author of the dataset justified that this dataset is based on 3 other datasets; the author may have removed or addressed instances during the integration process of creating this dataset.

2.2.2 Duplication Check

```
# Duplicate rows
print("Duplicate rows:", df_original.duplicated().sum())
```

✓ 1.7s

Duplicate rows: 2284

In the image above, the code shows that there are 2284 rows of duplicate entries in the data. For now, we are only identifying the duplicated rows, these rows will be revisited later on during the data cleansing stage to decide if they should be removed.

2.2.3 Unique Values Check

```
✓  
# Unique values in each column  
for column in df_original.columns:  
    print(f"{column}: {df_original[column].nunique()} unique values")  
2] ✓ 0.8s
```

The number of unique values were analysed for each column in order to recognise unexpected categories and unusual variable behaviour. Most of the categorical variables had small and consistent values. For example, gender contains 2 categories, and smoking, physical_activity, and stress_level each contain 3 categories. There were no apparent spelling errors, capitalisation, and formatting inconsistencies of category values.

The column 'composite_key' had a fairly low 192 unique values (for the total data set of 280,985 records). But this isn't too surprising because the column is a combination of characteristics instead of a unique identifier for each person.

The most unexpected discovery was with the 'heart_disease' column. The column seems like it would be a binary identifier representing whether or not an individual has heart disease. However it consists of 125 unique numerical values, including fractional values such as 0.08, 0.222222 and 0.032258. This shows that this column cannot be interpreted as a simple yes or no as we originally thought, and would require further interpretation in section 2.3.

```
composite_key: 192 unique values  
age_level: 4 unique values  
gender: 2 unique values  
bmi_level: 4 unique values  
smoking: 3 unique values  
diabetes: 2 unique values  
age: 88 unique values  
age_normalized: 151 unique values  
bmi: 14122 unique values  
hypertension: 137 unique values  
heart_disease: 125 unique values  
HbA1c_level: 192 unique values  
glucose: 154 unique values  
cholesterol: 323 unique values  
sleep_hours: 10092 unique values  
triglycerides: 525 unique values  
physical_activity: 3 unique values  
family_history: 2 unique values  
stress_level: 3 unique values  
low_hdl_cholesterol: 2 unique values  
high_ldl_cholesterol: 2 unique values  
blood_pressure: 197 unique values  
high_blood_pressure: 2 unique values  
sugar_consumption: 3 unique values  
crp_level: 9992 unique values  
...  
source_dataset: 3 unique values  
disease_flags: 8 unique values  
sublabel: 8 unique values  
label: 2 unique values  
Output is truncated. View as a scrollable element
```

2.2.4 Range Values

```
# Numerical ranges and summary statistics
df_original.describe().T
```

✓ 0.6s

	count	mean	std	min	25%	50%	75%	max
age	280985.0	49.665865	21.682350	2.000000	32.000000	50.000000	67.000000	89.000000
age_normalized	280985.0	0.547883	0.249222	0.000000	0.344827	0.551724	0.747126	1.000000
bmi	280985.0	27.490663	7.040373	10.010000	22.100000	27.320000	32.700000	95.690000
hypertension	280985.0	0.128218	0.187346	0.000000	0.000000	0.071429	0.217063	1.000000
heart_disease	280985.0	0.073390	0.138243	0.000000	0.000000	0.007299	0.115053	1.000000
HbA1c_level	280985.0	5.947475	0.954072	3.500000	5.384805	5.800000	6.865766	9.000000
glucose	280985.0	135.258646	38.477865	70.000000	100.000000	138.000000	160.000000	300.000000
cholesterol	280985.0	224.045525	35.525530	150.000000	202.000000	225.000000	243.000000	300.000000
sleep_hours	280985.0	6.954652	1.426412	4.000000	6.200000	6.975185	7.700000	10.000000
triglycerides	280985.0	178.995744	67.336085	50.000000	129.000000	179.000000	242.000000	400.000000
blood_pressure	280985.0	149.887200	4.148644	120.000000	148.946429	149.713320	150.720942	180.000000
crp_level	280985.0	7.489127	1.013898	0.003647	7.294863	7.458483	7.758562	14.997087
homocysteine_level	280985.0	12.465326	0.983375	5.000236	12.287763	12.488407	12.686175	19.999037
systolic_bp	280985.0	134.405105	20.542140	90.000000	126.000000	134.167413	143.000000	179.000000
diastolic_bp	280985.0	89.476720	13.667219	60.000000	84.000000	89.504587	95.000000	119.000000
alcohol_intake	280985.0	15.003336	6.830255	0.000000	12.000000	14.982818	17.900000	30.000000
salt_intake	280985.0	8.480577	2.960646	2.000000	7.200000	8.477600	9.800000	15.000000
heart_rate	280985.0	74.497343	11.402996	50.000000	70.000000	74.487247	79.000000	99.000000
hdl	280985.0	64.609147	15.938525	30.000000	58.000000	64.650655	72.000000	99.000000
ldl	280985.0	129.725919	27.352521	70.000000	118.000000	129.983051	141.000000	189.000000

The numerical values were analysed to show their maximum, median, average, and quartile values. Most column ranges shown above show plausible data. The columns cholesterol, glucose, hdl, ldl, triglycerides, and sleep hours all seemed to be reasonable. However there are some values which are unusual. One being a minimum value of 2, and the other is a maximum BMI of 95.69. These unusual values still cannot be considered errors from just their value, and we will need to analyse them more during data cleansing before any editing is done.

2.3 Data Interpretation and Assumptions

The initial inspection of the data revealed some important issues with how heart disease is represented in the data. The **heart_disease** column is represented as a range of values anywhere between 0 and 1, it is not a binary classification of true or false. The main goal of this project was to determine whether or not a person had a heart disease. This means that the **heart_disease** column cannot be used as the binary target.

This dataset also has a '**label**' column, this classifies each person under either 'normal' or 'abnormal', which is exactly the binary type we are looking for. However, 'abnormal' is way too vague and only identifies if a record is associated with a disease, it does not identify an exact disease.

The sublabel column provides a more precise classification for each record by assigning each record into the eight possible groupings of diseases; **N** for no listed disease, **DI** for diabetes, **HT** for heart disease, **HY** for hypertension, **DI-HT** for diabetes and heart disease, **DI-HY** for diabetes and hypertension, **HT-HY** for heart disease and hypertension, and **DI-HT-HY** for all three diseases. From these classifications, any sublabel that contains '**HT**' can be treated as a person having a heart disease. This is exactly the binary target we are looking for in this project, so the '**sublabel**' column will be used to construct a new binary heart disease target.

There are also several variables which represent similar data differently, for example **age**, **age_level**, and **age_normalized** all represent a person's age while **bmi** and **bmi_level** both represent the person's BMI. Later analysis will be carried out on these factors when deciding whether they are suitable to remain when modelling.

Disease related columns such as **sublabel**, **disease_flags**, **label** and others could directly reveal if there is a heart disease present. This means that they need to be removed from the predictor variables appropriately to prevent **data leakage**.

Finally, outlier values will not be automatically removed just because they are extreme. Instead they will first need to be investigated further during cleansing and analysis before deciding whether it should be kept, transformed or removed.

3. Data Cleansing and Transformation

3.1 Data Cleaning and Preparation

Explain the actual cleaning performed and justify the decisions. Include where relevant:

- ☐ missing-value treatment
- ☐ duplicate removal
- ☐ correction of invalid values
- ☐ category standardisation
- ☐ formatting corrections
- ☐ data-type conversion
- ☐ unit conversion
- ☐ reshaping or aggregation

3.2 Outlier Investigation and Treatment

Explain how potential outliers were investigated and whether they were retained, removed, capped, or transformed.

The decision should be justified rather than automatically treating every extreme value as an error.

3.3 Cleaned Dataset and Validation

Show that the cleaning process produced a valid dataset. Summarise:

- ☐ resulting dataset dimensions
- ☐ remaining missing values
- ☐ duplicate count
- ☐ corrected data types/ranges
- ☐ validation checks performed
- ☐ important differences between the raw and cleaned datasets

4. Exploratory Data Analysis

4.1 Descriptive and Distribution Analysis

Analyse the overall characteristics of the important variables. Include relevant:

- ☐ descriptive statistics
- ☐ numerical distributions
- ☐ categorical frequencies
- ☐ histograms
- ☐ boxplots
- ☐ bar charts
- ☐ target/class distribution
- ☐ class imbalance

4.2 Relationships and Patterns in the Data

Investigate relationships between variables and with the target. Include relevant:

- ☐ scatter plots
- ☐ grouped comparisons
- ☐ correlation analysis
- ☐ correlation heatmap
- ☐ relationships with the target
- ☐ missingness patterns
- ☐ unusual patterns or clusters

4.3 Key Findings and Dataset Implications

Bring the EDA together rather than ending with a collection of graphs. Summarise the most important findings about:

- ☐ dataset structure
- ☐ data quality
- ☐ important relationships
- ☐ imbalance
- ☐ unusual patterns
- ☐ apparent usefulness of the dataset for the intended problem
- ☐ issues that may need to be addressed later during modelling

5. Data Engineering Pipeline and Phase I Summary

5.1 Data Engineering Pipeline

Present the required pipeline diagram and briefly explain the overall process:

Raw Dataset → Acquisition → Inspection → Cleaning → Transformation → Validation → Cleaned Dataset → Exploratory Data Analysis

5.2 Phase I Summary and Next Steps

Briefly summarise what was achieved during Phase I and identify what the project will move into during Phase II, such as feature engineering and machine learning.

Heart Disease Target Note

- Heart disease is a broad category, not one specific condition. It can include coronary artery disease, heart failure, arrhythmias, valve disease, and cardiomyopathy.
- The dataset does not identify the specific type of heart disease.
- Therefore, the project should ask: *Can clinical and lifestyle measurements be used to predict whether a patient has heart disease?*
- Limitation: different heart conditions are grouped into one binary target.

References

1. Cardiovascular diseases are the leading cause of death globally (World Health Organization, 2025).
2. Telfar Barnard, L. (2025, July). Heart disease in Aotearoa: Morbidity, mortality, and service delivery. Prepared for Kia Manawanui Trust, University of Otago.